

Balboa Park Alive!

Exploring Biodiversity through Mobile Augmented Reality

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ABSTRACT

Balboa Park Alive! is a series of immersive, interactive phone-based installations that leverage Augmented Reality (AR) to explore biodiversity of San Diego/Tijuana. This application combines Niantic Lightship ARDK, Mapbox and mobile hand tracking in service of situated and embodied encounters with 3D digital renderings of some of the plants, insects, and animals that make this region one of the top biodiversity hotspots in the world. Unlike related conservation-centered mobile AR apps, *Balboa Park Alive!* prioritizes first-hand perspective taking and evidence-based methods of inquiry, using guided movement and visualizations to invite families to engage with their surroundings and each other.

KEYWORDS

Augmented reality, embodied cognition, mobile hand tracking

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1 INTRODUCTION

Balboa Park Alive! was conceived in response to the urgent global challenge of species loss due to environmental degradation. It centers on ecosystems of the San Diego/Tijuana region, which represents a biodiversity hotspot. Designed for families with children, the

app aims to foster interest in wildlife and support for conservation by promoting through imaginative, embodied perspective taking and situated learning. Mobile phone-based augmented reality (AR) is a well-suited medium for engaging both of these cognitive abilities – that is, perspective taking and learning – as it can deliver immersive content in authentic settings and transform opaque concepts around biological or ecological processes into playful and explorative visualizations. Harnessing these unique affordances of phone-based AR, the *Balboa Park Alive!* app prompts users to “see” the world from the perspective of a bee, to think like a scrub jay, or test hypotheses about plant/animal synergies while exploring the gardens of Balboa Park, a cultural hub of San Diego, California.

2 RELATED WORK

Over twenty years’ of existing work has advanced locative media designed to facilitate engagement with the outdoor physical environment. The Wearable Environmental Media platform (WEM) [Fisher 2001] was hailed as an early proof-of-concept of Bell Lab’s vision of a digitally mediated world-wide “communication skin” that would allow real-time information from diverse pervasive digital sensing systems to be accessed by users as they traversed their environments. It facilitated users’ exploration of green spaces on a university campus by allowing them to access location-specific multimedia content that constituted a “Virtual Field Guide” as they moved from place to place. One and a half decades later, the cultural sensation, Pokémon GO (Niantic Labs, 2016), expanded the notion of a “communication skin” to gaming. By integrating Global Positioning System (GPS), Visual Positioning System (VPS), and mobile AR capabilities, the game enabled users to “play” with virtual Pokémon in parks and other neighborhood locales. Since then, games have begun to emerge that leverage related location-based and AR technologies in service of conservation-based themes. For instance, *Rangers Wanted* (Theodore Roosevelt Presidential Library, 2024) tasks users with cultivating flowers for virtual bees.

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3 USER EXPERIENCE DESIGN AND THEORETICAL FOUNDATIONS

Like *Rangers Wanted*, *Balboa Park Alive!* aims to stimulate interest in and understanding of wildlife. Examples of experiences currently accessible through the *Balboa Park Alive!* app include an interactive honeybee waggle dance that guides visitors to the Zoro Garden (Figure 1, a-b), a pollinator vision modality that allows users to “see” the ultraviolet hues of flowers visible to some pollinators, and a planting and butterfly release game (Figure 1, c-d). However, in contrast to its counterparts, *Balboa Park Alive!* advances a new framework of situated, social, inquiry-based, and embodied learning. To support situated learning [Suchman 1987], installations were located in areas of the park where concepts visualized in the digital overlays can be readily mapped to features of the physical space, including existing plants, trees, and features of the terrain. To support embodied learning [Sharkey et al. 2022; Twomey et al. 2022], the activity design prioritized opportunities for reframing abstract concepts, such as UV perception, in terms of firsthand experience in which users either assume the role of an animal or organize their body movements in direct engagement with virtual plants and animals. Finally, to support social learning [Bandura and Walters 1977], *Balboa Park Alive!* offers families a shared activity that motivates social outdoor play and exploration.

Additionally, the app supports inquiry-based knowledge creation through activity designs that prompt users to generate and test hypotheses (e.g., Are spring azure butterflies attracted to the California lilac? Do hummingbirds prefer circular or tubular flowers?). If users require guidance or wish to find out more, they can consult a companion in-app e-book. Pages of the book that are relevant to a specific park site and topic can be accessed during engagement with that experience. This design was adopted to promote just in time learning through conversations between care givers and children centered on the book and immersive AR content.

4 SYSTEM DESIGN

Balboa Park Alive! incorporates Niantic Lightship tools, such as their ARDK, which is a comprehensive framework designed for AR applications. It integrates advanced AR functionalities like AR Camera, Local Positioning, and Object Detection, enabling accurate merging of digital and physical environments. This technology ensures that augmented objects are realistically placed within users’ views, enhancing the immersive nature of the AR experiences provided by the app (Niantic Labs, 2021). The integration of Mapbox SDK offers robust mapping and navigation tools that support real-time, precise geolocation features. This is crucial for guiding users through Balboa Park’s diverse ecosystems, providing them with dynamically updated navigational data that enhances their exploratory experience in the park (Mapbox, 2021). MediaPipe Hands is utilized for its real-time hand-tracking capabilities, facilitating natural interactions within the AR environment. This system interprets user hand movements, enabling interactions such as gesturing to plant seeds or follow bees.

5 CONCLUSION AND FUTURE STEPS

Balboa Park is the largest urban cultural park in the United States, attracting between 13 and 20 million visitors each year. Because



Figure 1: The *Balboa Park Alive!* AR app. a,b) Users navigate to the Zoro garden by interpreting a bee waggle dance superimposed over their hand. c,d) A family engages in a pollinator matching game.

phone-based AR readily affords both presence and shared social experience, the *Balboa Park Alive!* phone app is expected to bring thousands of families and friends together, kindling conversation, play, and outdoor exploration centered on biodiversity, ecology, and conservation. Initial user testing indicates that users, particularly families with children, find the AR experience engaging and informative. In the longer term, the app will be expanded to encompass more species and ecological interactions, and more advanced hand tracking and gesture recognition technologies will be explored to improve user interaction and engagement.

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